Potential Energy-Constrained Quantum Hybrid Attention Network for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) aims to identify pixels that differ from complex background distributions without prior target information. Deep learning methods such as convolutional neural networks (CNNs) and transformers have been widely investigated for HAD. However, complex nonlinear spectral relationships in highly mixed backgrounds remain difficult to characterize. Meanwhile, without prior information, training may lack guidance for separating backgrounds and anomalies, weakening anomaly–background separability and causing anomaly contamination. To address these issues, in this article, a potential energy-constrained quantum hybrid attention network (PEQA) is proposed for HAD with three key modules. Specifically, the quantum spectral–spatial attention (QSA) module embeds a quantum neural network (QNN) into the spectral attention branch, where spectral features are encoded into quantum states and channelwise similarity is measured by quantum fidelity, enabling nonlinear spectral interactions to be captured within the attention process. Moreover, the potential energy (PE) loss constrains bottleneck features from an energy-distribution perspective, allowing dense background-dominated latent regions to have lower potential values while making relatively isolated anomaly related samples more distinguishable. Finally, the background prior guidance (BPG) module provides auxiliary background guidance, with superpixel segmentation first generating locally homogeneous regions and singular value decomposition (SVD) extracting dominant components for background prior construction. The experimental results on six datasets indicate that PEQA achieves favorable detection performance compared with nine representative HAD methods.**

***Index Terms*—Anomaly detection, hyperspectral images, potential energy (PE) constraint, quantum neural networks (QNNs).**

I. Introduction

HYPERSPECTRAL imaging (HSI) technology captures electromagnetic radiation across continuous and narrow spectral bands, producing a 3-D data cube that contains rich spatial and spectral information [1]. By providing an almost continuous spectral signature for each spatial pixel, hyperspectral data enable subtle material differences to be distinguished, which are often difficult to identify from RGB or multispectral images [2]. Due to these advantages, hyperspectral data have been widely applied in diverse remote sensing scenarios, such as classification [3], change detection [4], and target detection [5].

As an important branch of hyperspectral data analysis, hyperspectral anomaly detection (HAD) aims to identify target pixels whose spectral signatures deviate significantly from the surrounding local or global backgrounds [6]. Unlike supervised target detection [7], HAD is generally performed without prior knowledge of anomalous targets or background land covers [8]. This characteristic is particularly valuable in practical remote sensing scenarios, where accurate spectral libraries and target labels are often difficult to obtain [9]. Given that anomalous targets usually occupy only a small fraction of a scene and occur with low probability, the key challenge of HAD lies in accurately characterizing complex background distributions, so that anomalous targets can be effectively highlighted as outliers [10].

Over the past decades, HAD methods have evolved from statistical background modeling to representation learning and deep feature extraction. Statistical methods generally identify anomalies by measuring deviations from the estimated background distribution [11]. Subsequently, representation-based methods were developed to exploit the spectral redundancy and spatial correlation of hyperspectral data, aiming to separate sparse anomalous components from the dominant background structure [12]. With the development of deep learning, HAD has gradually shifted toward data-driven nonlinear feature modeling [13]. Deep-learning-based methods generally learn background-dominant representations from hyperspectral data in an unsupervised manner and identify anomalies according to their deviations from the learned background [14]. These methods have enriched the technical framework of HAD and improved its adaptability to complex hyperspectral scenes [15].

Despite the progress achieved by deep-learning-based methods, HAD still faces several practical challenges. First, real hyperspectral scenes often contain complex backgrounds whose spectral signatures can be similar to those of anomalous targets [16]. In such cases, convolutional neural network (CNN) and attention-based feature extractors that rely mainly

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on classical linear projections or conventional similarity measures may not sufficiently amplify weak nonlinear spectral differences, making subtle anomalies easy to confuse with background pixels [17]. Second, many existing methods adopt the pixel-averaged mean squared error (mse) as the training objective [13]. However, this objective treats all pixels in a uniform manner and has limited ability to distinguish background pixels from anomaly related deviations that should be preserved for detection. As the loss is averaged over all pixels, the optimization mainly reduces the overall reconstruction error rather than enhancing anomaly–background separability, making weak anomalous deviations less distinguishable in the residual map [18]. Finally, HAD is an unsupervised task, where model training is directly conducted on the test image. Since the image contains both background pixels and unknown anomalous pixels, directly feeding the image into the network may introduce anomalous interference into the learning process [19]. Such interference can weaken the separation between background and anomalies and cause anomaly contamination [20].

To enhance nonlinear spectral modeling in HAD, quantum neural networks (QNNs) provide a useful perspective. In QNNs, classical features can be encoded into quantum states in a complex Hilbert space, where parameterized quantum transformations enable flexible interactions among input variables through quantum state evolution [21]. Such a mechanism is suitable for modeling nonlinear spectral relationships in HSIs, particularly in scenarios with diverse background variations and complex anomaly related spectral signatures. Recently, a QNN-based technique has also been introduced into HAD, where the complex quantum space is employed for feature aggregation and decision-making with a lightweight architecture [22]. This recent attempt suggests that QNNs can serve as a feasible tool for nonlinear information modeling in HAD.

Based on the above considerations, a potential energy-constrained quantum hybrid attention network (PEQA) is developed for HAD. First, to enhance nonlinear dependence modeling along the spectral dimension, a quantum spectral–spatial attention (QSA) module is designed. By incorporating QNN and quantum fidelity into spectral attention, spectral relationships can be modeled in a quantum feature space, providing a more flexible mechanism for nonlinear spectral correlation learning. Second, to overcome the limitation of pixel-averaged reconstruction objectives, a potential energy loss (PE Loss) is imposed on bottleneck features. Conventional mse-based optimization tends to uniformly reduce pixelwise reconstruction errors, which can lower the residual contrast of weak anomalies. PE Loss regularizes bottleneck features through a PE formulation, encouraging background-like pixels to stay in low-potential regions and increasing the energy gap for pixels that deviate from the background distribution. Finally, to mitigate background contamination during reconstruction, a background prior guidance (BPG) mechanism is incorporated before deep feature learning. Since anomalies generally occupy only a small portion of the scene, local spatial–spectral homogeneous regions can be used to estimate background-dominant information through superpixel-based grouping. In this way, BPG provides early background guidance for the reconstruction network and reduces the influence of anomalous pixels being directly learned during reconstruction. The main contributions of this article can be summarized as follows.

1) A QSA module is developed to enhance nonlinear spectral–spatial feature modeling. In the spectral attention branch, QNN is embedded to encode spectral features into quantum states, and quantum fidelity computed from quantum wave functions is used to reformulate spectral affinity estimation in the attention mechanism. By coupling quantum spectral attention with spatial interaction, QSA provides a more flexible nonlinear modeling mechanism for spectral–spatial dependencies.
2) A PE-constrained optimization strategy, implemented as PE Loss, is introduced for unsupervised reconstruction. In addition to the global reconstruction loss, PE Loss imposes an energy-distribution constraint on bottleneck features by formulating their latent PE. This design encourages background-like pixels to stay in low-potential regions while enlarging the energy gap for pixels deviating from the background distribution, thereby improving the preservation of weak anomalous pixels in residual-based detection.
3) A BPG module is developed to alleviate background contamination in reconstruction. Specifically, simple linear iterative clustering (SLIC) is used for structure-aware local grouping, and singular value decomposition (SVD) is adopted to extract low-rank background-dominant components before deep feature learning. By introducing local background guidance at an early stage, BPG reduces the influence of anomalous pixels being directly fit during reconstruction.

The remainder of this article is organized as follows. Section II reviews the related HAD methods, as well as the fundamentals of QNNs. Section III presents the proposed PEQA framework. Section IV provides the experimental results and analysis. Section V concludes this article.

II. Related Works

This section reviews representative studies related to HAD from three main aspects. First, traditional HAD methods are introduced, including statistical model-based and representation-based approaches. Then, deep-learning-based methods are discussed, covering autoencoder-based reconstruction, attention-based models, and other emerging network structures. Finally, QNN-related studies are reviewed to introduce their recent applications in remote sensing and hyperspectral image processing.

A. Traditional HAD Methods

Traditional HAD methods generally identify anomalous pixels by constructing explicit background descriptions and designing corresponding anomaly scores. Statistical model-based methods are a representative line of early HAD

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studies. The classical RX [23] detector estimates the background distribution with a Gaussian model and uses the Mahalanobis distance to measure the deviation of each pixel from the background. To improve its applicability in complex scenes, several RX-based extensions have been developed, such as local RX [24], kernel RX [25], and FrFE-RX [26]. These methods improve the original detector from different aspects, including local background estimation, nonlinear distribution modeling, sample weighting, and covariance regularization [27].

Representation-based methods constitute another important branch of traditional HAD. Their basic idea is that background pixels can be represented by neighboring samples, dictionaries, or low-dimensional structures, while anomalies usually lead to larger representation errors. Typical methods include sparse representation detector (SRD) [28], collaborative representation detector (CRD) [29], and low-rank representation [30]. In particular, low-rank and SRD [31] model the background as a low-rank component and separate anomalies as sparse residuals. Recent studies have further extended this line by incorporating more flexible priors and representation forms, such as mixture-of-Gaussian sparse modeling [32], generalized nonconvex surrogated tensor modeling for blurred HSIs [33], and feedback band grouping with variation low-rank sparse modeling [34]. However, these traditional methods usually depend on specific statistical assumptions or sensitive parameter settings, which may limit their robustness and generalization in complex hyperspectral scenes [35].

B. Deep-Learning-Based HAD Methods

Since HAD is performed without labeled anomaly samples, most deep-learning-based methods follow an unsupervised reconstruction strategy. The background is expected to be reconstructed by the network, while anomalies are highlighted by the residual between the input image and the reconstructed output.

As a representative branch of deep-learning-based HAD, AE-based methods usually reconstruct the background and locate anomalies according to reconstruction residuals. Wang et al. [36] introduced Auto_AD, which adopts a fully convolutional autoencoder with adaptive-weighted reconstruction for autonomous background estimation. Following this paradigm, various AE-based models have been developed, including guided autoencoder [37], robust graph autoencoder [38], and hyperbolic space-based autoencoder [39]. To alleviate the identity mapping problem in self-supervised reconstruction, Liu et al. [40], [41] proposed BiGSeT and MSNet by introducing separation training and further combining it with multiscale convolutional encoder–decoder layers. More recently, Cui et al. [42] proposed SuperAD to reduce anomaly reconstruction through superpixel pooling, error-adaptive convolution, and online background pixel mining. These studies indicate that AE-based reconstruction models have become an effective framework for background estimation in unsupervised HAD.

Attention-based methods have recently become an important direction in deep-learning-based HAD. Attention mechanisms can strengthen spectral–spatial feature interaction and long-range dependence modeling. Wu and Wang [43] proposed TAEF, which adopts a Transformer encoder to capture local-to-global spatial information and embeds a nonlinear mixing model into the decoder for background reconstruction. Lian et al. [11] developed GT-HAD, where a gated Transformer uses content similarity to separately model background and anomaly features during reconstruction. In addition, finite spatialwise attention-based self-supervised detectors [44], global-to-local spatial–spectral awareness Transformer networks [45], and dual-decoder dual-Transformer reconstruction networks [46] have also been investigated for HAD. These methods further improve the representation capability of deep HAD models by introducing spectral–spatial dependence modeling.

In addition to AE and attention-based methods, other deep learning models have also been explored for HAD, including blind-spot networks [47], frequency-domain networks [15], and Mamba-based reconstruction networks [48]. Overall, deep-learning-based methods reduce the dependence on handcrafted assumptions and improve automatic feature learning for HAD. However, under the unsupervised reconstruction paradigm, background contamination and insufficient spectral discrimination remain challenging issues.

C. Quantum Neural Networks

QNNs provide a hybrid learning framework by combining quantum computing with classical neural networks. In general, classical inputs are encoded into quantum states through qubits and quantum gates, processed by parameterized quantum circuits (PQCs), and then measured as classical outputs for optimization [49]. Benefiting from quantum superposition and entanglement, QNNs can characterize complex feature interactions in a high-dimensional Hilbert space, which makes them potentially useful for modeling nonlinear and uncertain data distributions.

QNNs have also been explored in image analysis and Earth observation, where quantum or hybrid quantum–classical circuits are usually integrated into classical networks to enhance feature encoding and representation learning. In remote sensing, circuit-based hybrid QNNs and quantum–classical convolutional networks have been introduced for scene classification and multispectral land-cover classification [50]. For hyperspectral data, Lin et al. [51] proposed a quantum information-empowered graph neural network for hyperspectral change detection, where quantum information is incorporated into graph-based feature learning. Recently, Lv et al. [52] proposed QSCDNet for hyperspectral image change detection, where a quantum spectral change branch based on PQCs is combined with a classical CNN branch to model spectral change uncertainty and extract change-sensitive features. Although QNNs have shown potential in remote sensing and hyperspectral image processing, their use in HAD is still rarely explored.

III. Methodology

The overall architecture of the proposed PEQA is shown in Fig. 1, where an encoder–decoder framework is adopted

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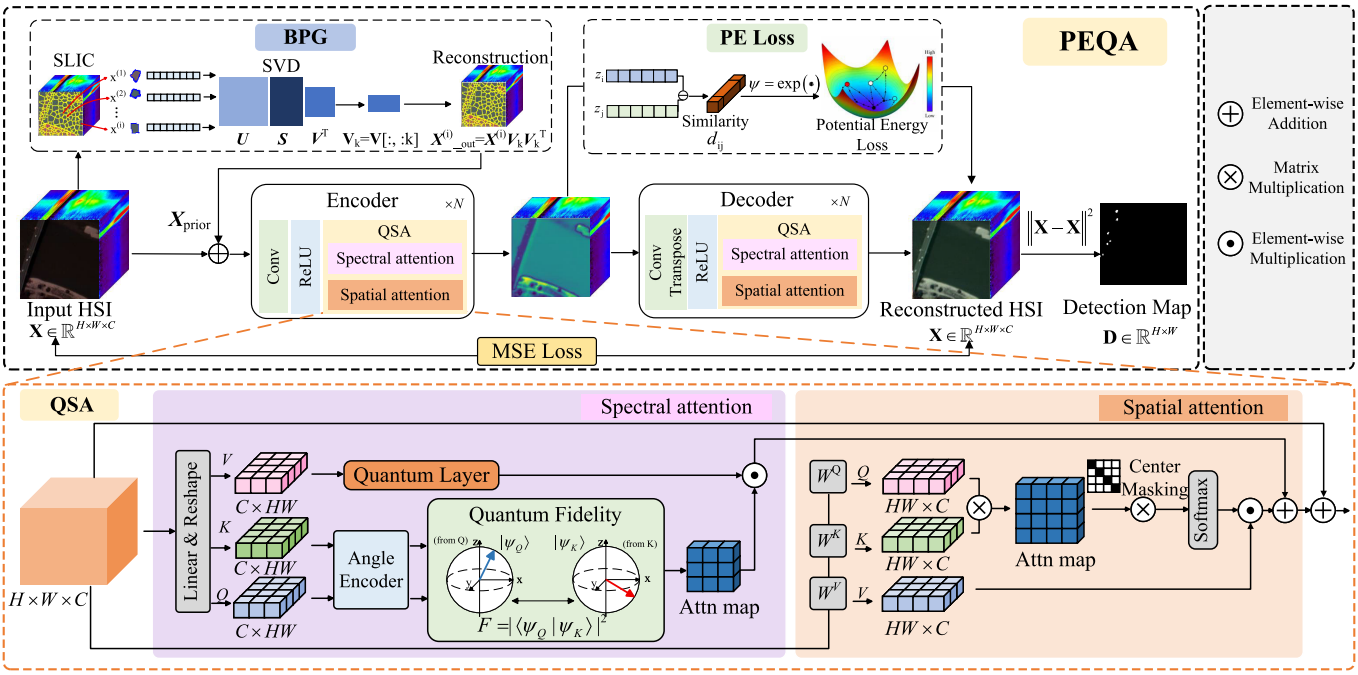


Fig. 1. Overall framework of the proposed PEQA. The input HSI is first processed by the BPG branch to generate a background prior, which is added to the original input and then fed into the encoder–decoder network. QSA modules are inserted in both the encoder and decoder stages, and (bottom) enlarged view shows the internal data flow of the QSA module, including quantum-fidelity-based spectral attention and spatial attention. The PE Loss is computed from the bottleneck features, while the mse loss is applied between the input and reconstructed HSI. Finally, the detection map is obtained from the reconstruction residual between the input HSI and the reconstructed HSI.

for HAD. Given an input hyperspectral image, a BPG branch is first introduced to extract background prior information based on SLIC and SVD. The obtained prior is then fused with the original input and forwarded to the encoder for feature extraction. To enhance spectral–spatial representation, a QSA module is embedded into both the encoder and decoder stages. Finally, the detection map is generated under the supervision of the proposed PE Loss together with other objective terms. The details of the BPG branch, QSA module, and optimization strategy are presented in Sections III-A–III-C.

A. Background Prior Guidance

The BPG branch aims to provide a reliable background prior for the subsequent anomaly detection process. Given that anomalous targets are usually sparsely distributed in the scene, SLIC is first utilized to impose local spatial homogenization by grouping spectrally similar neighboring pixels into superpixels. This operation reduces the disturbance caused by small anomalous regions and helps extract locally consistent background structures.

Let $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ be the input hyperspectral image, where H, W, and C represent the spatial height, spatial width, and number of spectral bands, respectively. As shown in Fig. 1, SLIC first partitions $\mathbf{X}$ into N_s nonoverlapping superpixel regions, which can be denoted as

$$\mathbf{X} \xrightarrow{\text{SLIC}} \left\{x^{(1)}, x^{(2)}, \ldots, x^{(N_s)}\right\}, \quad \mathbf{X} = \bigcup_{i=1}^{N_s} x^{(i)} \tag{1}$$

where $x^{(i)}$ denotes the ith superpixel region. Here, N_s is a hyperparameter that controls the number of generated superpixels. A larger N_s produces finer local regions, while a smaller N_s leads to coarser spatial grouping.

After the SLIC-based local grouping, SVD is further performed within each superpixel to extract the dominant spectral–spatial components. Since background pixels usually occupy most local regions and exhibit strong spectral redundancy, the dominant components mainly describe the background distribution, while sparse anomaly related perturbations can be suppressed during low-rank approximation. In this way, a refined background prior is obtained before feature encoding, which helps alleviate anomaly contamination and increase the distinction between background and anomalies.

For each superpixel $x^{(i)}$, its pixels are rearranged into a 2-D matrix form, where each row corresponds to one pixel and each column corresponds to one spectral band. SVD is then applied as

$$x^{(i)} = U^{(i)} S^{(i)} \left(V^{(i)}\right)^T \tag{2}$$

where $U^{(i)}$ and $V^{(i)}$ are the left and right singular matrices, respectively, and $S^{(i)}$ is a diagonal matrix whose entries are singular values sorted in descending order. Since the dominant background information is mainly concentrated in the principal singular components, only the first k singular components are retained to construct the truncated subspace

$$V_k^{(i)} = \left[v_1^{(i)}, v_2^{(i)}, \ldots, v_k^{(i)}\right] \tag{3}$$

where k is a hyperparameter that determines the rank of the low-rank reconstruction.

For the ith superpixel, let n_i be the number of pixels in this region. After rearrangement, $x^{(i)}$ is represented as a matrix

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$x^{(i)} \in \mathbb{R}^{n_i \times C}$, where each row corresponds to one pixel and each column corresponds to one spectral band. Accordingly, in the compact SVD form, $U^{(i)} \in \mathbb{R}^{n_i \times r_i}$, $S^{(i)} \in \mathbb{R}^{r_i \times r_i}$, and $V^{(i)} \in \mathbb{R}^{C \times r_i}$, where $r_i = \min(n_i, C)$. The retained matrices are $U_k^{(i)} \in \mathbb{R}^{n_i \times k}$, $S_k^{(i)} \in \mathbb{R}^{k \times k}$, and $V_k^{(i)} \in \mathbb{R}^{C \times k}$, with $k \le r_i$. Based on the retained principal subspace, the reconstructed background prior of the ith superpixel is obtained by

$$\hat{x}^{(i)} = \left(x^{(i)} V_k^{(i)}\right)\left(V_k^{(i)}\right)^T = U_k^{(i)} S_k^{(i)} \left(V_k^{(i)}\right)^T \tag{4}$$

where $U_k^{(i)}$ and $S_k^{(i)}$ represent the top k left singular vectors and the corresponding principal singular values, respectively. It can be seen that $x^{(i)} V_k^{(i)} \in \mathbb{R}^{n_i \times k}$ and $\hat{x}^{(i)} \in \mathbb{R}^{n_i \times C}$, which has the same dimension as the original superpixel matrix. Since the columns of $V_k^{(i)}$ are orthonormal right singular vectors, $x^{(i)} V_k^{(i)} = U_k^{(i)} S_k^{(i)}$, which ensures the equivalence of the two reconstruction forms in (4). Through this truncated reconstruction, the major background structure can be preserved, while sparse anomaly related perturbations are relatively suppressed.

After all superpixels are processed, the reconstructed regions $\{\hat{x}^{(1)}, \hat{x}^{(2)}, \ldots, \hat{x}^{(N_s)}\}$ are reassembled according to their original spatial positions, yielding the complete background prior map $\mathbf{X}_{prior} \in \mathbb{R}^{H \times W \times C}$. Finally, the extracted background prior $\mathbf{X}_{prior}$ is fused with the raw input $\mathbf{X}$ through elementwise addition, and the fused result is fed into the encoder for subsequent feature extraction

$$\mathbf{X}_{fused} = \mathbf{X} + \mathbf{X}_{prior}. \tag{5}$$

By introducing an explicit background reference before feature encoding, the influence of anomalous pixels can be reduced at an early stage, thereby providing more reliable background guidance for the reconstruction network.

B. Quantum Spectral–Spatial Attention

Given an intermediate feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$ from the encoder or decoder, a convolution layer followed by a ReLU activation is first used for local feature refinement. The refined feature is then projected and reshaped into query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ tensors. In the proposed QSA module, the spectral branch focuses on channelwise correlation modeling, while the spatial branch further captures local spatial interactions.

1) Spectral Branch: Recent quantum-related studies have introduced QNNs into HAD, indicating that quantum circuits can provide an alternative nonlinear mapping mechanism for hyperspectral feature processing [22]. Different from directly using QNN as an independent detector or decision module, the proposed QSA embeds a lightweight QNN into the spectral attention branch, where the QNN is used to refine the value tensor and quantum fidelity is further adopted to estimate channelwise spectral affinity.

Given the value tensor $\mathbf{V} \in \mathbb{R}^{C \times HW}$, a classical projection layer first maps each spectral feature vector into a C_q-dimensional quantum input space, where C_q denotes the number of qubits. Then, angle encoding is used to prepare the input quantum state

$$\left|\psi_{\text{in}}\right\rangle = \bigotimes_{i=1}^{C_q} R_Y\left(\tilde{v}_i\right)\left|0\right\rangle. \tag{6}$$

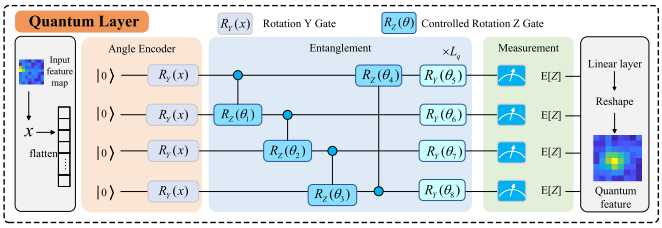


Fig. 2. Architecture of the QNN module. The input feature vector is first projected into a C_q-dimensional quantum input, encoded by R_Y gates, processed through L_q parameterized entanglement layers, and measured with Pauli-Z observables. The measured values are then mapped back to form the quantum-refined feature used in the spectral attention branch.

In (6), $|\psi_{\text{in}}\rangle$ denotes the input quantum state represented by the Dirac ket notation, where ψ_{in} is a state vector in the quantum Hilbert space. Each projected feature value $\tilde{v}_i$ is used as the rotation angle of the ith qubit. The rotation gate $R_Y(\alpha)$ is implemented according to its standard matrix form

$$R_Y(\alpha) = \begin{bmatrix} \cos(\alpha/2) & -\sin(\alpha/2) \\ \sin(\alpha/2) & \cos(\alpha/2) \end{bmatrix}. \tag{7}$$

Therefore, each initial state $|0\rangle$ is rotated into a feature-dependent quantum state by $R_Y(\tilde{v}_i)$. The tensor product operator $\otimes$ combines all C_q encoded qubits into a multiqubit quantum state, which is then fed into the subsequent PQC.

The encoded quantum state is further processed by a PQC composed of trainable rotation gates and entanglement operations

$$\left|\psi_{\text{out}}\right\rangle = U(\boldsymbol{\theta})\left|\psi_{\text{in}}\right\rangle \tag{8}$$

where $U(\boldsymbol{\theta})$ denotes the trainable unitary transformation implemented by the PQC, and $\boldsymbol{\theta}$ represents the trainable circuit parameters, including the rotation angles in the quantum gates. As illustrated in Fig. 2, this compact quantum processing flow includes classical projection, angle encoding, parameterized quantum transformation, quantum measurement, and output projection.

The circuit output is converted back to classical features by measuring the expectation value of the Pauli-Z observable on each qubit

$$z_i = \left\langle \psi_{\text{out}} \middle| \sigma_z^{(i)} \middle| \psi_{\text{out}} \right\rangle, \quad i = 1, 2, \ldots, C_q. \tag{9}$$

Here, $\sigma_z^{(i)}$ denotes the Pauli-Z measurement operator acting on the ith qubit, and z_i is the corresponding expectation value used as the measured quantum feature.

The measured vector is then projected back to the original channel dimension, producing the quantum-refined value tensor $\mathbf{V}_q \in \mathbb{R}^{C \times HW}$. In this way, the QNN introduces a compact nonlinear transformation into the value branch, while the following fidelity-based attention calculation determines the spectral correlation weights.

While $\mathbf{V}_q$ is refined by the QNN, the spectral attention weights are computed from the query and key tensors using a quantum-fidelity-based similarity measure. The query and key features are first encoded into single-qubit states through angle encoding. For the cth query channel and the c'th key channel

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at the nth spatial position, the corresponding quantum states are defined as

$$\left|q_{c,n}\right\rangle = R_{\mathrm{Y}}\left(\phi_{c,n}^{\mathrm{Q}}\right)\left|0\right\rangle \quad \left|k_{c',n}\right\rangle = R_{\mathrm{Y}}\left(\phi_{c',n}^{\mathrm{K}}\right)\left|0\right\rangle \tag{10}$$

where $\phi_{c,n}^{\mathrm{Q}}$ and $\phi_{c',n}^{\mathrm{K}}$ denote the rotation angles encoded from the query and key features, respectively.

The similarity between the two encoded quantum states is measured by quantum fidelity

$$\mathcal{F}_{c,c',n} = \left|\left\langle q_{c,n}\middle|k_{c',n}\right\rangle\right|^2 = \cos^2\left(\frac{\phi_{c,n}^{\mathrm{Q}} - \phi_{c',n}^{\mathrm{K}}}{2}\right). \tag{11}$$

Compared with conventional dot-product similarity, quantum fidelity measures the overlap between two encoded quantum states. Therefore, it provides a nonlinear spectral affinity measure in the attention process, which is helpful for characterizing weak spectral differences between anomalies and complex backgrounds.

For spectral attention calculation, each channelwise query or key feature is regarded as a sequence with length $N = HW$. The spectral similarity between the cth query channel and the c'th key channel is obtained by accumulating the logarithmic fidelity scores over all spatial positions

$$\mathbf{S}_{\mathrm{spec}}\left(c, c'\right) = \sum_{n=1}^{N} \log\ \left(\mathcal{F}_{c,c',n} + \varepsilon\right) \tag{12}$$

where ε is a small constant for numerical stability. The logarithmic form converts the product of fidelity scores into a summation form, reducing numerical instability when many spatial positions are involved.

The spectral attention map is then generated by applying a rowwise Softmax operation to the scaled similarity matrix

$$\mathbf{A}_{\mathrm{spec}} = \mathrm{Softmax}\left(\frac{\mathbf{S}_{\mathrm{spec}}}{\sqrt{N}}\right). \tag{13}$$

Finally, the output of the spectral branch is obtained by multiplying the spectral attention map with the quantum-refined value tensor

$$\mathbf{X}_{\mathrm{spectral}} = \mathbf{A}_{\mathrm{spec}}\mathbf{V}_q. \tag{14}$$

2) Spatial Branch: Parallel to the quantum spectral analysis, the spatial query ($\mathbf{Q}_s$), key ($\mathbf{K}_s$), and value ($\mathbf{V}_s$) tensors are projected from the output of the spectral branch using linear layers.

The spatial interaction is then modeled by a scaled dot-product attention mechanism. To reduce direct self-reconstruction and encourage each spatial position to aggregate contextual information from other positions, a center-masking strategy is incorporated into the attention calculation. Specifically, after the spatial feature map is flattened into $N = HW$ positions, a mask matrix $\mathbf{M} \in \mathbb{R}^{N\times N}$ is constructed as

$$M_{ij} = \begin{cases} -\infty, & i = j \\ 0, & i \neq j \end{cases} \tag{15}$$

where the diagonal entries correspond to the attention between each position and itself. By assigning $-\infty$ to the diagonal entries before the Softmax operation, the self-attention responses of the center positions are suppressed, while the correlations with other spatial positions are preserved. In practical implementation, $-\infty$ is replaced by a sufficiently large negative value for numerical stability. Overall, this operation prevents each spatial position from directly attending to itself and encourages the spatial branch to aggregate contextual responses from other positions, thereby reducing the risk of direct self-reconstruction in the reconstruction-based anomaly detection framework. The spatially refined feature representation is computed as

$$\mathbf{X}_{spatial} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_s\mathbf{K}_s^{\top}}{\sqrt{d}} + \mathbf{M}\right)\mathbf{V}_s. \tag{16}$$

Finally, the outputs of the spectral and spatial branches are fused through elementwise addition. The final output feature is given by

$$\mathbf{X}_{out} = \mathbf{X}_{spatial} + \mathbf{X}_{spectral}. \tag{17}$$

Through this design, local spatial structures and quantum-enhanced spectral features are jointly integrated into a unified representation.

C. PE Constraint

The PE constraint is introduced to regularize bottleneck features from an energy-distribution perspective. In the bottleneck feature space, background-related samples usually form dense and compact distributions, which are expected to correspond to lower potential values, whereas anomaly related samples tend to deviate from the dominant distribution and are more likely to be associated with higher potential values. Based on this characteristic, PE Loss is imposed on the bottleneck feature to encourage dominant background features to form compact low-potential regions and increase the potential separation between background and anomaly related features.

Let the bottleneck feature be denoted by $\mathbf{Z} \in \mathbb{R}^{H_z\times W_z\times C_z}$, where H_z, W_z, and C_z denote the height, width, and channel number of the bottleneck feature, respectively. After reshaping, $\mathbf{Z}$ is converted into a set of latent vectors $\{\mathbf{z}_i\}_{i=1}^{N}$, where $N = H_z \times W_z$. Each spatial location in the bottleneck feature map is thus treated as one latent sample for potential modeling.

For the latent samples, the pairwise squared Euclidean distance is defined as

$$d_{ij}^2 = \left\|\mathbf{z}_i - \mathbf{z}_j\right\|_2^2. \tag{18}$$

Based on this distance, a Gaussian interaction term is introduced as

$$\psi_{ij} = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \tag{19}$$

where σ controls the effective neighborhood scale and is adaptively set as the median pairwise distance

$$\sigma = \mathrm{median}_{i<j}\left(\left\|\mathbf{z}_i - \mathbf{z}_j\right\|_2\right). \tag{20}$$

This median-based scale is less affected by sparse anomaly related samples and mainly reflects the neighborhood scale of the dominant background distribution.

The Gaussian term ψ_{ij} measures the local affinity between latent samples $\mathbf{z}_i$ and $\mathbf{z}_j$, assigning larger weights to nearby samples and smaller weights to distant ones. Accordingly, the

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accumulation of ψ_{ij} over neighboring samples can be used to characterize the local density concentration in the latent space.

The local density around the ith latent sample is estimated by accumulating the contributions from all latent samples

$$\rho(\mathbf{z}_i) = \sum_{j=1}^{N} \psi_{ij} = \sum_{j=1}^{N} \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_j\|_2^2}{2\sigma^2}\right) \tag{21}$$

where the reference sample itself is also retained in the summation $\sum_{j=1}^{N} \psi_{ij}$; the case of $j = i$ is included. For the self-interaction case, $d_{ii} = 0$ and $\psi_{ii} = \exp(0) = 1$. Therefore, the density estimate can be equivalently decomposed as $\rho(\mathbf{z}_i) = 1 + \sum_{j \neq i} \psi_{ij}$. The first term is a constant baseline contributed by the reference sample itself, while the sample-dependent density variation is still determined by the affinities between $\mathbf{z}_i$ and the other latent samples.

A density-weighted dispersion term is further introduced as

$$s(\mathbf{z}_i) = \sum_{j=1}^{N} d_{ij}^2 \psi_{ij} = \sum_{j=1}^{N} \|\mathbf{z}_i - \mathbf{z}_j\|_2^2 \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_j\|_2^2}{2\sigma^2}\right). \tag{22}$$

Different from the density term, the self-interaction term does not contribute to the density-weighted dispersion because $d_{ii}^2 \psi_{ii} = 0$. Thus, $s(\mathbf{z}_i)$ is equivalently determined by $\sum_{j \neq i} d_{ij}^2 \psi_{ij}$, indicating that the dispersion term only reflects the weighted distances between $\mathbf{z}_i$ and other latent samples.

Based on the above two terms, the potential associated with $\mathbf{z}_i$ is defined as

$$V(\mathbf{z}_i) = \frac{s(\mathbf{z}_i)}{\rho(\mathbf{z}_i) + \varepsilon} = \frac{\sum_{j=1}^{N} \|\mathbf{z}_i - \mathbf{z}_j\|_2^2 \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_j\|_2^2}{2\sigma^2}\right)}{\sum_{j=1}^{N} \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_j\|_2^2}{2\sigma^2}\right) + \varepsilon} \tag{23}$$

where ε is a small constant introduced for numerical stability. Therefore, after the self-interaction term is included in $V(\mathbf{z}_i)$, it only adds a constant normalization contribution to the denominator, while the numerator remains unchanged. The relative potential differences are still determined by the nonself interactions with neighboring latent samples.

In the above formulation, a latent sample located in a dense and compact region tends to have a relatively small potential value, while an isolated sample in a sparse region is more likely to produce a larger potential value.

To provide an intuitive interpretation of the defined potential values, Fig. 3 visualizes the potential distribution on the Pavia dataset. Fig. 3(a) maps $V(\mathbf{z}_i)$ back to the spatial image domain, where most homogeneous background regions correspond to low-potential areas, while several locally heterogeneous or target-related positions present higher potential values. Fig. 3(b) further compares the potential-value distributions of background and anomaly pixels. The anomaly distribution is shifted toward larger potential values, and its mean value is higher than that of the background distribution. This visualization indicates that the potential formulation is consistent with the desired latent organization, in which dominant background features are assigned lower potential values and anomaly related features retain relatively higher potential values.

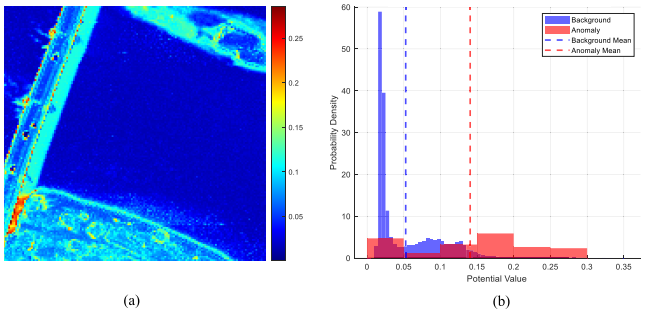


Fig. 3. Visualization of the latent potential values on the Pavia dataset. (a) Spatial potential map obtained by mapping $V(\mathbf{z}_i)$ back to the image domain, where brighter colors indicate higher potential values. (b) Probability distributions of $V(\mathbf{z}_i)$ for background and anomaly pixels, where dashed lines denote their mean potential values. The right-shifted anomaly distribution and its larger mean value indicate that anomaly related samples generally have higher potential values than background samples.

Accordingly, the proposed PE Loss is formulated as the average potential over all latent samples

$$\mathcal{L}_{\mathrm{PE}} = \frac{1}{N} \sum_{i=1}^{N} V(\mathbf{z}_i). \tag{24}$$

Minimizing $\mathcal{L}_{\mathrm{PE}}$ encourages dominant background features to form compact low-potential regions in the latent space. In contrast, anomaly related features usually deviate from the dominant background distribution and tend to retain relatively higher potential values, making them easier to distinguish during residual-based detection.

The reconstruction term $\mathcal{L}_{rec}$ is defined as the mse between the reconstructed output $\hat{\mathbf{X}}$ and the input hyperspectral image $\mathbf{X}$

$$\mathcal{L}_{rec} = \frac{1}{HWC} \left\|\hat{\mathbf{X}} - \mathbf{X}\right\|_F^2. \tag{25}$$

Thus, the overall training objective is written as

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_p \mathcal{L}_{PE} \tag{26}$$

where λ_p is the balance coefficient of the PE constraint.

After network optimization, the anomaly detection map is generated from the reconstruction residual between the input HSI and the reconstructed HSI. Let $\mathbf{x}_{i,j}$ and $\hat{\mathbf{x}}_{i,j}$ be the spectral vectors of $\mathbf{X}$ and $\hat{\mathbf{X}}$ at spatial position (i, j), respectively. The anomaly score is calculated as

$$\mathbf{D}(i, j) = \frac{1}{C} \left\|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\right\|_2^2 \tag{27}$$

where $\mathbf{D} \in \mathbb{R}^{H \times W}$ denotes the final detection map.

IV. Experiments and Results Analysis

This section presents a comprehensive evaluation of the proposed PEQA method through comparisons with nine representative HAD methods on six hyperspectral datasets. The experimental datasets are first introduced, followed by the qualitative and quantitative metrics adopted for performance assessment. The comparative results are then analyzed from visual quality, ROC-based measurements, parameter sensitivity analysis, ablation study, and computational efficiency perspectives.

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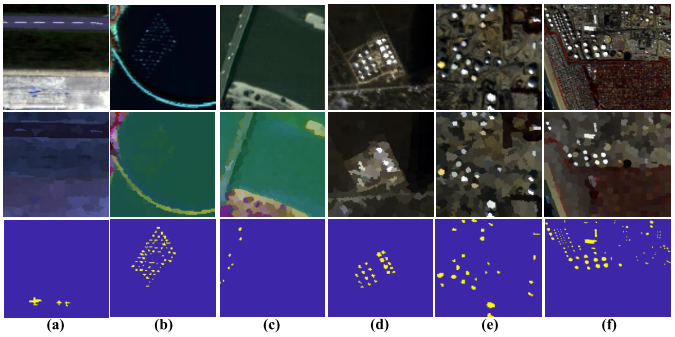


Fig. 4. Visualization of the six HSI datasets used in the experiments: (a) Gulfport, (b) Bay, (c) Pavia, (d) Coast, (e) Urban, and (f) Segundo. The first row shows the pseudo-color images, the second row shows the generated background prior $\mathbf{X}_{\text{prior}}$ by the BPG branch, and the third row shows the ground-truth maps.

A. Datasets Description

Six hyperspectral datasets are used to evaluate the detection performance of the proposed method. The pseudo-color images, the generated background priors $\mathbf{X}_{\text{prior}}$ by the BPG branch, and the ground-truth maps of these datasets are shown in Fig. 4. As shown in the second row of Fig. 4, the generated $\mathbf{X}_{\text{prior}}$ generally preserves the main background structures of different scenes, while the anomalous targets are relatively weakened. This provides a visual illustration of the output of the BPG branch. These datasets cover diverse scenes, including airports, beaches, urban areas, and industrial regions, with land cover types such as soil, water, roads, and residential areas. The anomalous targets mainly consist of aircraft, vehicles, industrial structures, and other man-made objects. For each dataset, the data size is reported in the form of $H \times W \times C$, where H and W denote the spatial height and width, respectively, and C denotes the number of spectral bands. The details of the six datasets are described as follows.

1) *Gulfport [53]:* Captured in Gulfport using the AVIRIS sensor, with a spatial resolution of 3.4 m/pixel and a size of $100 \times 100 \times 191$. This dataset contains 60 anomalies, mostly consisting of vehicles and aircraft.
2) *Bay [54]:* Captured in San Diego Bay using the AVIRIS sensor, with a spatial resolution of 7.5 m/pixel and a size of $150 \times 150 \times 193$. It includes 202 anomalies, which are primarily man-made objects on the beach.
3) *Pavia [55]:* Captured in Pavia using the ROSIS sensor, with a spatial resolution of 1.3 m/pixel and a size of $150 \times 150 \times 102$. It contains 68 anomalies, mainly corresponding to vehicles on the bridge.
4) *Coast [56]:* Captured in Texas Coast using the AVIRIS sensor, with a spatial resolution of 17.2 m/pixel and a size of $100 \times 100 \times 207$. It contains 67 anomalies, mostly urban structures.
5) *Urban [57]:* Captured in Los Angeles using the AVIRIS sensor, with a spatial resolution of 7.1 m/pixel and a size of $100 \times 100 \times 205$. This dataset contains 232 anomalies, including vehicles and buildings.
6) *Segundo [58]:* Captured over the El Segundo area in California, USA, using the AVIRIS sensor. It has a spatial resolution of 7.1 m/pixel and a size of $250 \times 300 \times 224$. The dataset contains 224 spectral bands ranging from 366 to 2496 nm. This scene includes diverse facilities, such as a park, a petroleum refinery, residential areas, and a school district. The anomalies mainly correspond to industrial structures, including storage tanks and towers at the petroleum refinery.

These datasets contain different types of anomalous targets under varied spatial–spectral backgrounds, providing a diverse evaluation environment for HAD.

B. Experimental Setup

1) Comparison Methods and Parameter Settings: To evaluate the performance of the proposed PEQA method, nine representative HAD methods were selected for comparison, including RX [23], CRD [29], LRASR [31], Auto_AD [36], DCAE [59], MSNet [41], TAEF [43], SSHAD [60], and DPMN [61]. Among them, RX, CRD, and LRASR are traditional HAD methods, while Auto_AD, DCAE, MSNet, TAEF, SSHAD, and DPMN are deep-learning-based HAD methods.

The specific hyperparameter settings of all comparison methods are described as follows. RX does not require parameter tuning. For CRD, the inner and outer window sizes were set to 9 and 11, respectively. For LRASR, the numbers of clusters and basis vectors were set to 15 and 20, respectively, while the regularization coefficients β and λ were fixed at 0.1 and 1. For DCAE, the number of clusters was determined using the elbow method, and the model was trained for 200 epochs. For MSNet, the optimization process was conducted with five iterations, each containing 150 training epochs. For Auto_AD, the batch size was set to 64, the learning rate was set to 0.001, and the network was trained for 100 epochs. For TAEF, the spatial neighborhood size was set to 15×15, and the tradeoff parameter was set to 0.01. For SSHAD, the spatial patch size was set to 13×13, the hidden embedding dimension was set to 64, and the learning rate was set to 5e-4. For DPMN, the number of mixture components was set to 5, and the model was trained for 150 epochs.

For the proposed PEQA method, the Adam optimizer was adopted for network training. The initial learning rate was set to 1e-3, and the maximum number of training iterations was set to 200. The balance coefficient for the loss function was set to 1e-3. The number of quantum channels in the QNN was set to 4, and the network consists of two layers. The same parameter settings were used for all datasets unless otherwise specified.

All experiments were implemented on a PC with 16 GB of RAM, an Intel Core i9-13900HX CPU, and an NVIDIA GeForce RTX 4060 GPU.

2) Qualitative and Quantitative Evaluation Metrics: Both qualitative and quantitative metrics are adopted to assess the anomaly detection performance on the six datasets. For qualitative analysis, color anomaly maps are presented to visually compare the detection maps of different methods, so that the highlighting of anomalous targets and the suppression of background regions can be observed more directly.

For quantitative evaluation, the 2-D receiver operating characteristic (2D-ROC) curve [62], 3-D receiver operating

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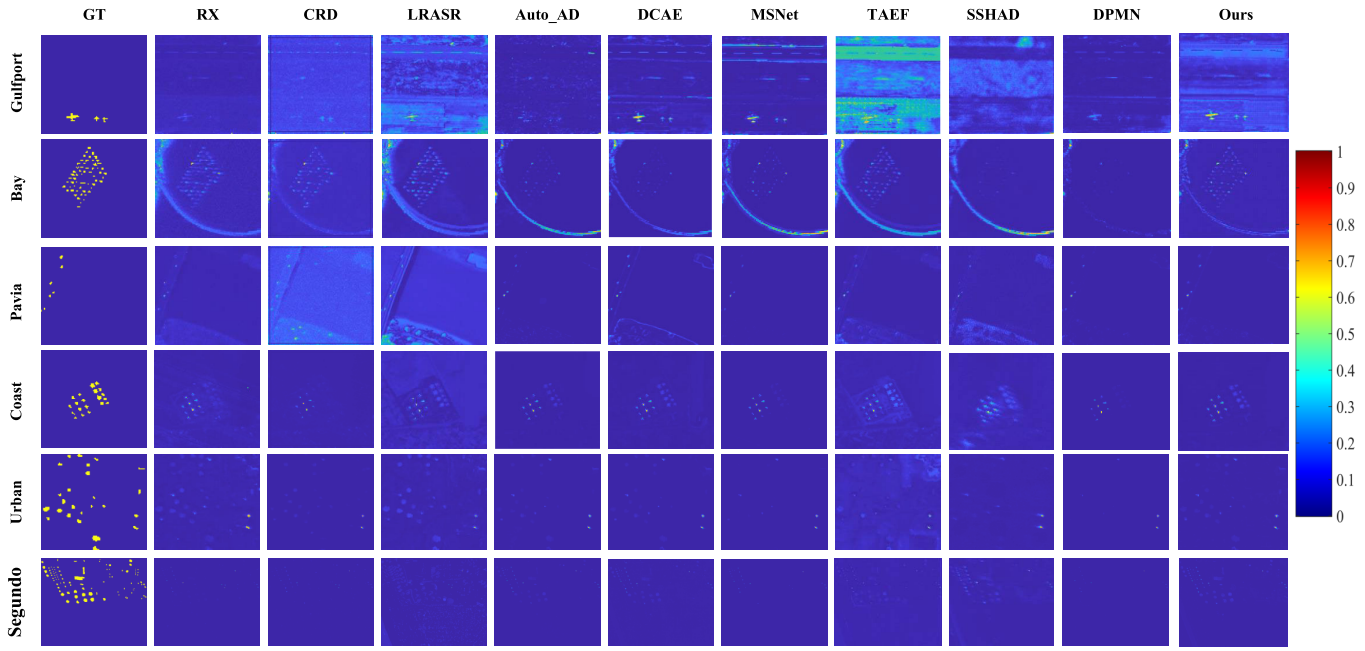


Fig. 5. Color anomaly maps based on ten methods for the six HSI datasets. All anomaly maps are normalized into the range of [0, 1] for visualization, where darker colors indicate lower anomaly responses and brighter colors indicate higher anomaly responses.

characteristic (3D-ROC) [63] curve, area under the curve (AUC), and box-whisker plots are employed. The 2D-ROC and 3D-ROC curves describe the relationships among detection probability P_D, false alarm probability P_F, and threshold τ. Specifically, three AUC metrics are adopted, including $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$. Among them, larger values of $\text{AUC}_{(D,F)}$ and $\text{AUC}_{(D,\tau)}$ indicate better detection performance, and the corresponding curves are expected to be closer to the upper left region. In contrast, a smaller value of $\text{AUC}_{(F,\tau)}$ indicates better background suppression ability, and the corresponding curve is expected to be closer to the lower right region. In addition, box-whisker plots are used to reflect the distributions of anomalous and background pixels, which further reveal their separability.

C. Comparison Results

In this section, the detection performance of all methods is analyzed qualitatively and quantitatively on the six datasets.

1) Qualitative Evaluation: The color anomaly maps of the ten methods are shown in Fig. 5. For the Gulfport scene, the anomaly targets are located in an airport background with runway-like and stripe-like structures. LRASR, TAEF, and SSHAD introduce obvious background activations in these structured regions, while DCAE and MSNet can highlight part of the target pixels but still retain horizontal interference. In comparison, PEQA provides clearer target indications and achieves a better balance between anomaly highlighting and background suppression, although slight stripe-like interference remains.

For the Bay and Pavia scenes, the backgrounds contain coastal textures, sand, water, and small sparse targets. In Bay, RX, LRASR, and TAEF retain noticeable arc-shaped background interference along the coast, whereas SSHAD and DPMN suppress the background more strongly but also weaken part of the anomaly cluster. PEQA preserves more target-related pixels with less coastal interference. In Pavia, CRD, LRASR, and SSHAD produce visible activations on diagonal or textured background regions, while PEQA keeps the background cleaner and maintains clearer indications for sparse anomaly pixels.

For the Coast and Urban scenes, the targets are embedded in complex urban or man-made structures. In Coast, LRASR, TAEF, and SSHAD activate many background pixels around structured areas, which may increase false alarms. In Urban, RX, LRASR, and TAEF also show scattered or broad background interference. By contrast, PEQA highlights the main target regions more clearly and suppresses most large-area background fluctuations, making the anomalous pixels easier to distinguish.

For the Segundo scene, the background is more heterogeneous because of industrial facilities, residential regions, and other spatial structures. LRASR and TAEF show visible background fluctuations, while several deep-learning-based methods generate relatively weak target contrast in some anomaly regions. PEQA produces a more stable anomaly map and retains more distinguishable target-related activations in the left-side regions of the scene. Overall, the visual comparisons indicate that PEQA can better preserve anomaly related regions while reducing structured background interference across different scenes.

2) Quantitative Evaluation: The quantitative results are reported in Table I, and the corresponding 2D-ROC and 3D-ROC curves are shown in Figs. 6 and 7, respectively. As listed in Table I, PEQA achieves the best $\text{AUC}_{(D,F)}$ values on all six datasets, indicating that higher detection probabilities can be maintained under different false alarm levels. For $\text{AUC}_{(D,\tau)}$, PEQA also obtains the best results on most datasets, which suggests that anomalous pixels can

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TABLE I
AUC VALUES OF DIFFERENT METHODS ON SIX HSI DATASETS

Dataset	Metric	RX	CRD	LRASR	Auto_AD	DCAE	MSNet	TAEF	SSHAD	DPMN	Ours
Gulfport	$AUC_{(D,F)}$	0.9526	0.8554	0.9552	0.9284	0.9962	0.9723	0.9545	0.2841	0.9730	**0.9984**
	$AUC_{(D,\tau)}$	0.0736	0.1798	0.4359	0.0772	0.4893	0.3815	0.6592	0.0447	0.1471	**0.7649**
	$AUC_{(F,\tau)}$	0.0248	0.1051	0.1394	0.0133	0.0186	0.0208	0.3233	0.0999	**0.0128**	0.0818
Bay	$AUC_{(D,F)}$	0.9106	0.9531	0.9064	0.8877	0.9131	0.8764	0.9073	0.7915	0.8490	**0.9588**
	$AUC_{(D,\tau)}$	0.1530	0.1493	0.1689	0.0357	0.0214	0.0472	0.1101	0.0209	0.0082	**0.2060**
	$AUC_{(F,\tau)}$	0.0488	0.0422	0.0432	0.0165	0.0101	0.0190	0.0326	0.0270	**0.0042**	0.0592
Pavia	$AUC_{(D,F)}$	0.9538	0.9440	0.8134	0.9767	0.9726	0.9848	0.9647	0.7523	0.9879	**0.9942**
	$AUC_{(D,\tau)}$	0.1343	**0.3071**	0.2454	0.1237	0.1798	0.0907	0.1367	0.0892	0.0987	0.2626
	$AUC_{(F,\tau)}$	0.0233	0.1467	0.0830	0.0017	0.0043	**0.0007**	0.0126	0.0200	0.0018	0.0066
Coast	$AUC_{(D,F)}$	0.9946	0.9398	0.9186	0.9944	0.9958	0.9968	0.9914	0.9753	0.9932	**0.9992**
	$AUC_{(D,\tau)}$	0.1177	0.0511	0.1262	0.0940	0.1068	0.0711	0.1706	0.1475	0.0484	**0.2834**
	$AUC_{(F,\tau)}$	0.0135	0.0077	0.0313	0.0021	0.0054	0.0005	0.0207	0.0212	**0.0004**	0.0128
Urban	$AUC_{(D,F)}$	0.9887	0.9336	0.9940	0.9914	0.9925	0.9895	0.8299	0.8295	0.9556	**0.9962**
	$AUC_{(D,\tau)}$	0.0891	0.0266	0.0827	0.0387	0.0409	0.0224	0.0704	0.0351	0.0157	**0.1255**
	$AUC_{(F,\tau)}$	0.0114	0.0017	0.0113	0.0010	0.0013	0.0002	0.0304	0.0084	**0.0001**	0.0099
Segundo	$AUC_{(D,F)}$	0.9841	0.8914	0.6848	0.9771	0.9848	0.9774	0.9040	0.8904	0.9132	**0.9931**
	$AUC_{(D,\tau)}$	0.0119	0.0106	0.0180	0.0225	0.0300	0.0082	0.0411	0.0406	0.0046	**0.0575**
	$AUC_{(F,\tau)}$	0.0017	0.0025	0.0060	0.0008	0.0052	**0.0001**	0.0134	0.0101	0.0003	0.0046

Note: For $AUC_{(D,F)}$ and $AUC_{(D,\tau)}$, higher values indicate better performance, whereas lower $AUC_{(F,\tau)}$ values are preferred. The best and second-best results for each metric are highlighted in bold and underlined, respectively.

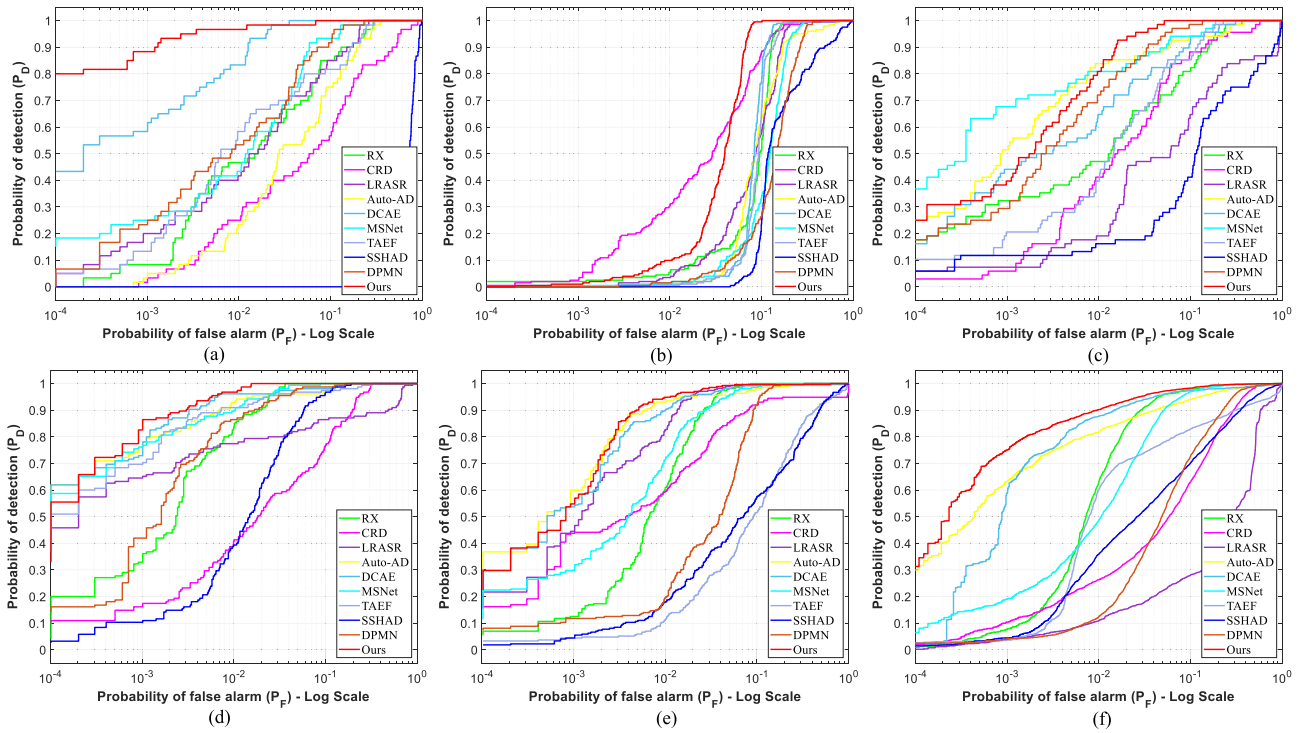


Fig. 6. ROC curves based on ten methods for the six HSI datasets. (a) Gulfport. (b) Bay. (c) Pavia. (d) Coast. (e) Urban. (f) Segundo.

be assigned relatively stable detection scores over a broad threshold range. These two metrics indicate that PEQA is effective in highlighting anomalous pixels.

In contrast, the ranking of PEQA on $AUC_{(F,\tau)}$ is relatively lower, since this metric mainly measures the false alarm probability along the threshold axis, and a smaller value is preferred. This result may be explained by the fact that PEQA is designed to preserve sufficient detection intensity for anomalous pixels rather than indiscriminately suppressing all high-score regions. In complex scenes, several background pixels with structures or spectra similar to anomalies may also retain moderate score values, leading to a less dominant advantage in $AUC_{(F,\tau)}$.

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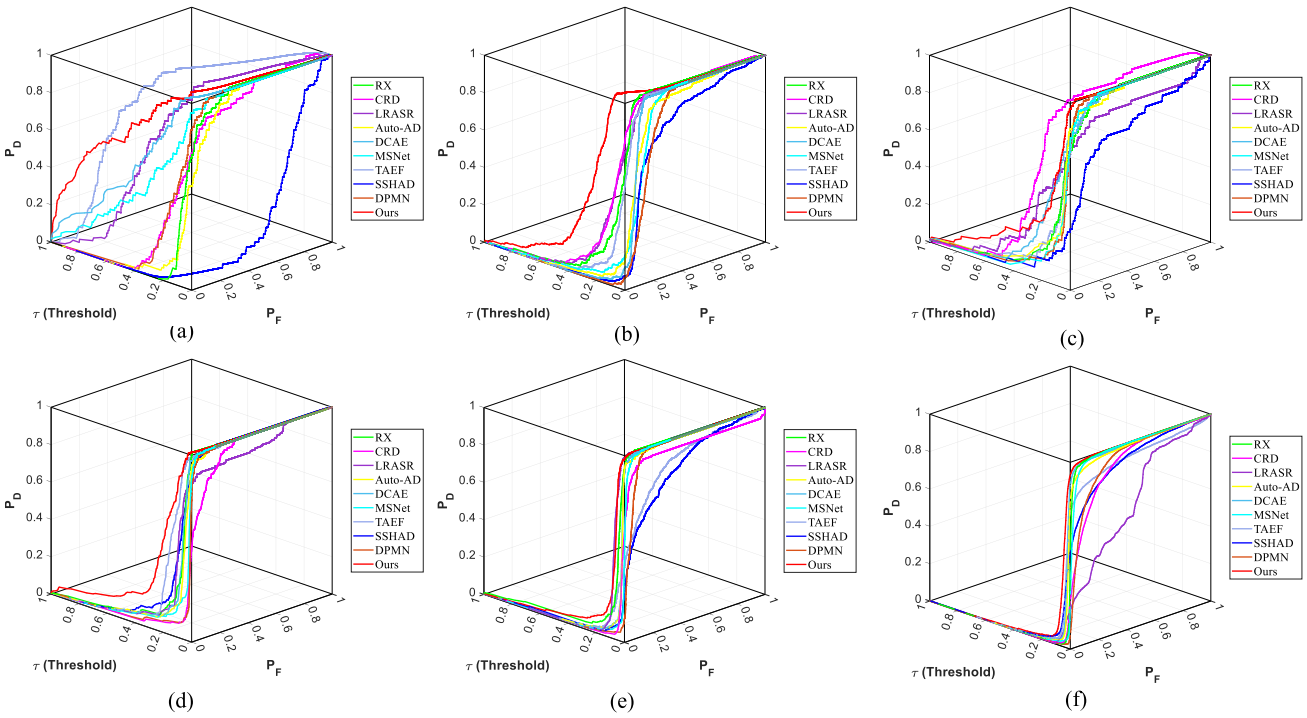


Fig. 7. 3D-ROC curves based on ten methods for the six HSI datasets. (a) Gulfport. (b) Bay. (c) Pavia. (d) Coast. (e) Urban. (f) Segundo.

Therefore, the quantitative results reveal a practical tradeoff: PEQA obtains stronger anomaly detection stability, while its background suppression is not always the most aggressive among all comparison methods.

The 2D-ROC curves in Fig. 6 are consistent with the AUC results. In most scenes, the curve of PEQA is closer to the upper left region, especially at relatively low false alarm levels, which indicates a favorable balance between detection probability and false alarm control. The 3D-ROC curves in Fig. 7 further reflect the influence of threshold variation. PEQA generally maintains high detection accuracy across a wide threshold range, while the false-alarm-related behavior is more moderate rather than consistently minimal. This observation agrees with the different rankings of $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{(D,\tau)}$, and $\mathrm{AUC}_{(F,\tau)}$ in Table I.

Fig. 8 shows the box-whisker plots of anomaly and background detection scores. A larger distance between the anomaly and background score distributions indicates better separability, while a compact background distribution reflects stronger false alarm suppression. Compared with most competing methods, PEQA generally produces a clearer interval between anomaly and background scores, with anomalous pixels concentrated at higher score levels. Although some background pixels still obtain relatively high scores in complex scenes, the overall distribution separation is more favorable, which further supports the observations from the AUC values and ROC curves.

In summary, the qualitative maps, ROC curves, AUC values, and box-whisker plots consistently support the effectiveness of PEQA. The proposed method improves anomaly–background separability while maintaining a reasonable balance between anomaly enhancement and background suppression. This result illustrates the complementary roles of BPG, QSA, and PE-constrained optimization.

D. Parameter Analysis

In this section, the sensitivity of PEQA to five key parameters is analyzed, including the PE Loss weight λ_p, the number of SLIC superpixels N_s, the SVD rank k, the number of quantum channels C_q, and the number of entanglement layers L_q. During the analysis, only one parameter is changed each time, while the remaining parameters are fixed to their default settings. The $\mathrm{AUC}_{(D,F)}$ results on six datasets are shown in Fig. 9.

As shown in Fig. 9(a), λ_p is selected from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. When λ_p is 10^{-5} or 10^{-4}, the PE constraint is relatively weak, which limits its regularization effect on bottleneck features. When λ_p increases to 10^{-2} or 10^{-1}, the latent constraint becomes stronger and may affect reconstruction learning. Among the tested values, $\lambda_p = 10^{-3}$ achieves stable performance on the six datasets and is adopted as the default setting.

For the BPG-related parameters in Fig. 9(b) and (c), N_s is selected from $\{50, 100, 150, 200, 250, 300\}$, and k is selected from $\{1, 3, 5, 7, 9, 11\}$. A small N_s, such as 50 or 100, may merge heterogeneous regions into the same superpixel, whereas a large N_s, such as 250 or 300, may divide local background regions into overly small parts. Similarly, a small k, such as 1 or 3, may remove useful background information, while a large k, such as 9 or 11, may retain more anomaly related interference. Based on the overall results, $N_s = 200$ and $k = 7$ are used to obtain the background prior.

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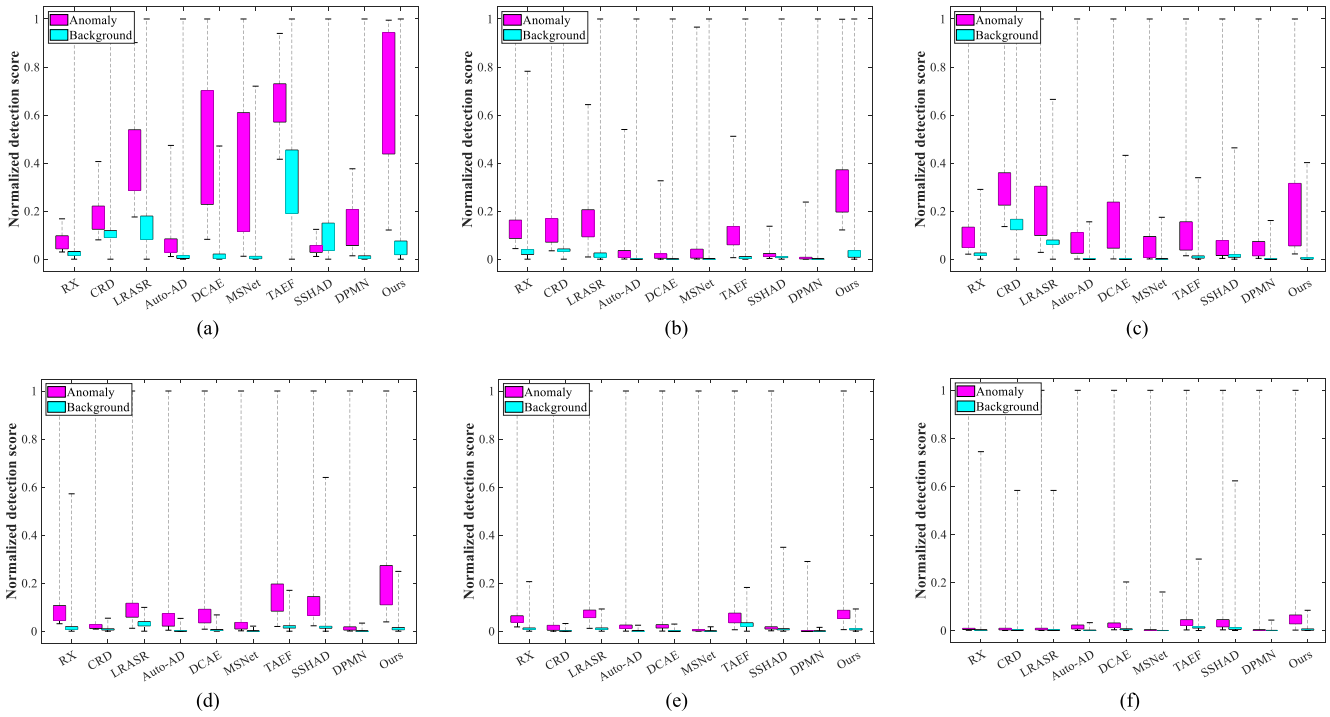


Fig. 8. Box-whisker plots based on ten methods for the six HSI datasets. (a) Gulfport. (b) Bay. (c) Pavia. (d) Coast. (e) Urban. (f) Segundo.

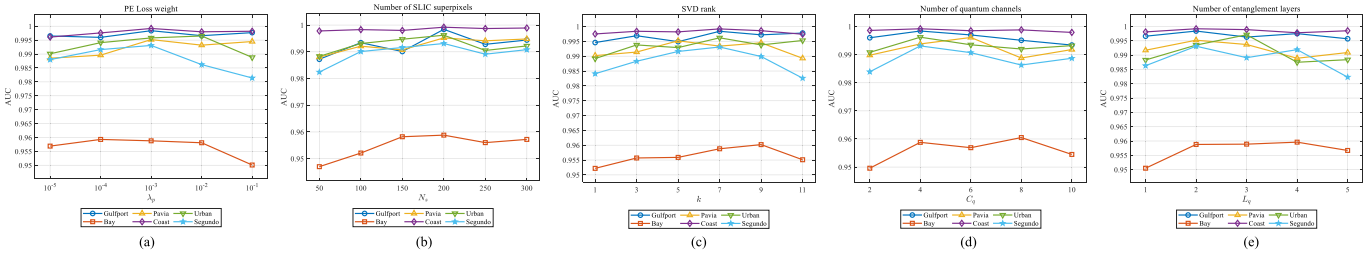


Fig. 9. Parameter sensitivity analysis of PEQA on six datasets. (a) PE Loss weight λ_p. (b) Number of SLIC superpixels N_s. (c) SVD rank k. (d) Number of quantum channels C_q. (e) Number of entanglement layers L_q.

For the QNN-related parameters in Fig. 9(d) and (e), C_q is selected from $\{2, 4, 6, 8, 10\}$, and L_q is selected from $\{1, 2, 3, 4, 5\}$. When $C_q = 2$, the quantum feature encoding capacity is relatively limited. Increasing C_q to 4 provides a more effective representation for spectral interaction modeling, whereas further increasing C_q to 6, 8, or 10 does not consistently improve the detection accuracy and also increases the cost of quantum encoding and measurement. Therefore, $C_q = 4$ is selected by considering both accuracy and training efficiency. For L_q, one entanglement layer provides a relatively shallow quantum transformation, while $L_q = 2$ improves the interaction among encoded spectral features. Further increasing L_q to 3, 4, or 5 introduces more trainable quantum operations and longer training time, but does not bring consistent performance gains. Therefore, $L_q = 2$ is adopted in the final implementation.

In summary, PEQA maintains stable detection performance within the tested parameter ranges. The selected settings, i.e., $\lambda_p = 10^{-3}$, $N_s = 200$, $k = 7$, $C_q = 4$, and $L_q = 2$, are used for all datasets unless otherwise specified.

E. Computational Complexity Analysis

In this section, the computational complexity of the proposed PEQA method is analyzed in terms of execution time, parameter count, and floating-point operations (FLOPs) [64]. For the proposed PEQA method, the reported running time in Table II includes the complete processing pipeline, including SLIC-based superpixel segmentation and SVD-based background prior construction. These metrics provide a quantitative reference for evaluating the computational burden of different methods. However, execution time, parameter count, and FLOPs are not strictly positively correlated. The actual running time can be affected by algorithmic implementation, hardware configuration, parallel computing efficiency, and data preprocessing strategies. In addition, some methods require extra preprocessing before detection, while the corresponding preprocessing time is not included in the reported results. Therefore, the values in Table II should be regarded as a relative comparison under the same experimental setting rather than an absolute measurement of practical deployment efficiency.

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TABLE II
COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT METHODS ON SIX HSI DATASETS

Dataset	Metric	RX	CRD	LRASR	Auto_AD	DCAE	MSNet	TAEF	SSHAD	DPMN	Ours
Gulfport	Time(s)	**0.1801**	5.2835	16.7917	7.5598	20.2136	10.7647	102.6524	28.8559	20.9478	42.4052
	Param(M)	–	–	–	3.2304	0.2429	**0.0929**	0.2282	1.0071	0.8731	0.8446
	Flops(G)	–	–	–	11.8174	4.8563	0.9224	**0.0228**	3.9262	6.3477	5.9838
Bay	Time(s)	**0.1923**	5.2386	20.0136	6.1098	12.1349	11.8081	84.7297	32.4108	20.7747	46.3953
	Param(M)	–	–	–	3.2332	0.2446	**0.0929**	0.2329	1.0079	0.8802	0.8470
	Flops(G)	–	–	–	11.8392	4.8909	0.9224	**0.0232**	3.9494	6.4192	6.0082
Pavia	Time(s)	**0.0330**	10.8504	28.4394	7.2884	32.8783	25.8321	133.4138	29.3768	23.2771	56.4359
	Param(M)	–	–	–	3.1050	0.0783	0.0929	**0.0674**	0.9552	0.6830	0.7354
	Flops(G)	–	–	–	24.4143	3.5184	2.0751	**0.0067**	6.4425	9.2312	11.2050
Coast	Time(s)	**0.0352**	5.1569	10.5329	4.3168	14.8554	14.2840	81.7767	28.6227	20.2482	38.6868
	Param(M)	–	–	–	3.2529	0.2819	**0.0929**	0.2672	1.0160	0.8428	0.8642
	Flops(G)	–	–	–	11.9915	5.6353	0.9224	**0.0267**	4.1118	6.5638	6.1796
Urban	Time(s)	**0.1872**	5.1670	12.3147	4.7394	12.0432	13.6207	87.0942	25.8189	21.1160	44.0798
	Param(M)	–	–	–	3.2501	0.2737	**0.0929**	0.2622	1.0149	0.8948	0.8617
	Flops(G)	–	–	–	11.9698	5.4720	0.9224	**0.0262**	4.0886	6.7385	6.1551
Segundo	Time(s)	**1.2107**	38.7121	125.1251	21.7881	113.2065	85.3842	396.5757	42.6189	58.6379	147.2428
	Param(M)	–	–	–	3.2769	0.3247	**0.0929**	0.3121	1.0259	0.8796	0.8850
	Flops(G)	–	–	–	91.2028	48.6912	6.9186	**0.0311**	32.2478	53.2848	48.0940

Note: For each efficiency metric, smaller values indicate better computational efficiency. The best and second-best results among the available values are highlighted in bold and underlined, respectively. "–" indicates that the metric is not applicable.

As shown in Table II, traditional methods such as RX and CRD generally require less execution time, especially RX, which achieves the fastest running speed on all six datasets. This is mainly because these methods have simple computational procedures and do not involve trainable network parameters. Nevertheless, their lightweight structures may limit their ability to model complex spatial–spectral distributions. LRASR also has no network parameters, but its running time is relatively high on several datasets due to the iterative optimization process. For deep-learning-based methods, Auto_AD shows relatively short execution time on some datasets, but it has larger parameter counts and FLOPs. DCAE and MSNet have compact parameter sizes, while their runtime varies across datasets. TAEF has the lowest FLOPs among the learning-based methods, but its execution time remains relatively long, indicating that FLOPs alone cannot fully reflect actual computational efficiency. Compared with SSHAD and DPMN, PEQA has a moderate parameter scale, although its execution time is not the shortest. The relatively longer execution time of PEQA may be attributed to the QSA module, where quantum-fidelity-based attention calculation and QNN operations introduce extra computations for spatial–spectral feature modeling.

Overall, PEQA achieves a reasonable balance between detection performance and computational efficiency. Although it is not the fastest method and does not have the smallest parameter count or FLOPs, its running time remains within an acceptable range considering its detection accuracy. The comparison also indicates that computational efficiency should not be evaluated by a single metric alone, since a method with fewer FLOPs or parameters may still require longer execution time in practice. Therefore, PEQA provides a practical tradeoff between computational cost and anomaly detection performance, making it suitable for scenarios where detection accuracy is more important than extreme computational speed.

TABLE III
ABLATION STUDY OF DIFFERENT COMPONENTS

QSA	PE Loss	BPG	Gulfport	Bay	Pavia	Coast	Urban	Segundo
			0.9824	0.9041	0.9718	0.9949	0.9887	0.9808
✓			0.9965	0.9465	0.9919	0.9970	0.9900	0.9884
	✓		0.9939	0.9199	0.9875	0.9961	0.9930	0.9918
		✓	0.9877	0.9055	0.9879	0.9985	0.9951	0.9911
✓	✓		0.9973	0.9496	0.9848	0.9983	0.9949	0.9918
✓		✓	0.9917	0.9368	0.9934	0.9990	0.9932	0.9895
	✓	✓	0.9914	0.9188	0.9916	0.9987	0.9927	0.9863
✓	✓	✓	**0.9984**	**0.9588**	**0.9942**	**0.9992**	**0.9962**	**0.9931**

Note: The best and second-best results among the available values are highlighted in bold and underlined, respectively.

F. Ablation Study

To evaluate the contribution of each component in PEQA, ablation experiments are conducted by selectively enabling QSA, PE Loss, and BPG. The $AUC_{(D,F)}$ results are reported in Table III. The first row corresponds to the baseline encoder–decoder reconstruction network without the three components, while the last row denotes the full PEQA model.

1) Effectiveness of QSA: QSA is the primary component of PEQA and brings consistent improvements over the baseline on all six datasets. The gains are especially clear on Bay, Pavia, and Segundo, where complex spectral–spatial variations and weak target cues make anomaly detection more difficult. This indicates that quantum-fidelity-based spectral interaction,

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together with spatial attention, is helpful for improving the spectral–spatial feature modeling capability of the reconstruction network. Compared with the baseline model, QSA makes anomalous pixels more distinguishable under heterogeneous background conditions, thereby providing a stronger basis for residual-based detection.

2) Effectiveness of PE Loss: PE Loss also improves the baseline performance on all datasets when used alone. The improvement is relatively clear on Pavia, Urban, and Segundo, suggesting that the latent-space energy constraint helps separate dominant background features from sparse anomaly related features. Different from modules that modify the network architecture, PE Loss regularizes the feature distribution during optimization. Thus, it improves the discrimination of reconstruction residuals without introducing an additional inference branch.

3) Effectiveness of BPG: BPG further improves the baseline on all datasets, although its single-module gain is relatively moderate in some scenes such as Bay. This is reasonable because BPG mainly provides a background-oriented reference before feature encoding, and its effect depends on whether a reliable background prior can be extracted from the input scene. For datasets with more structured or low-rank background characteristics, such as Coast, Urban, and Segundo, BPG brings more noticeable improvements. These results indicate that introducing a background prior is beneficial for reducing anomalous interference during reconstruction.

When two components are combined, the results are not strictly monotonic on every dataset. Some two-component variants improve the baseline clearly, but may be slightly inferior to a single component in specific scenes. This phenomenon suggests that QSA, PE Loss, and BPG affect the reconstruction process from different perspectives, and a simple pairwise combination may cause a local imbalance between target enhancement and background suppression. Nevertheless, the full PEQA model achieves the best performance on all six datasets. This suggests that the three components are complementary: QSA strengthens spectral–spatial feature interaction, PE Loss improves latent-space separability, and BPG provides additional background guidance. Their joint use leads to more stable and effective detection performance across different scenes.

V. Conclusion

In this article, a PEQA is proposed for HAD. The proposed framework consists of three connected components. First, the QSA module combines quantum state encoding, fidelity-based spectral correlation measurement, and spatial contextual interaction to enhance nonlinear spectral–spatial feature modeling. Then, the PE Loss introduces an energy-distribution constraint during network optimization, encouraging background-dominant pixels to maintain lower potential values while increasing the potential difference between background and anomalous pixels. Finally, the BPG module is introduced before deep feature learning. By using SLIC-based local grouping and SVD-based low-rank reconstruction, BPG provides an auxiliary background reference to reduce anomaly contamination during reconstruction. The experimental results on six hyperspectral image datasets indicate that PEQA achieves competitive detection performance compared with nine representative HAD methods.

Nevertheless, PEQA still has room for improvement in computational efficiency. The QNN-based spectral attention involves quantum-state-related computation, and the computational cost may increase when deeper entanglement layers are used or large-scale hyperspectral images are processed. Future work will focus on accelerating QNN computation and improving the implementation efficiency of PEQA.

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